
THE ASSOCIATION BETWEEN SEDENTARY BEHAVIOR, PHYSICAL ACTIVITY, AND SLEEP DISTURBANCE: A MEDIATION ANALYSIS OF C-REACTIVE PROTEIN

[ANONYMIZED FOR REVIEW]

ABSTRACT

Objective: Sedentary behavior and physical activity are thought to influence sleep via inflammatory mechanisms, but no study has directly tested whether inflammatory biomarkers mediate these associations. We tested whether log-transformed high-sensitivity C-reactive protein (hs-CRP) mediated the cross-sectional associations between daily sedentary time, moderate-to-vigorous physical activity (MVPA), and sleep disturbance in a large community sample. **Methods:** We analyzed data from 2,481 adults ($M_{\text{age}} = 45.0$ years, 50.4% female) in the 2005–2006 National Health and Nutrition Examination Survey (NHANES). Sedentary time and MVPA were self-reported; sleep disturbance was assessed via a validated composite proxy; hs-CRP was assayed from fasting blood. A parallel mediation model (PROCESS Model 4) with 5,000 bootstrap resamples tested indirect effects, controlling for age, sex, BMI, smoking, alcohol, depression, education, income, race/ethnicity, and comorbidity. **Results:** None of the seven pre-registered hypotheses were supported. Contrary to predictions, greater sedentary time was associated with lower sleep disturbance ($b = -0.002$, $p = .007$, $f^2 = 0.003$). MVPA was unrelated to sleep disturbance ($b = 0.001$, $p = .141$). Hs-CRP was not associated with either behavior or sleep (all $ps > .23$), and neither indirect effect was significant (both 95% CIs included zero). Depression was the dominant predictor of sleep disturbance (full model $R^2 = .19$). **Conclusions:** The hypothesized mediation pathway was not supported. Inflammatory biomarkers likely do not mediate behavior–sleep associations in cross-sectional community data. Future work should examine alternative mechanisms and use longitudinal designs with objective activity measurement.

Keywords sleep disturbance; C-reactive protein; sedentary behavior; physical activity; mediation analysis

Sleep disturbance affects approximately one in three adults in the United States and is a transdiagnostic risk factor for cardiovascular disease, metabolic dysfunction, depression, and all-cause mortality (?). Two modifiable behavioral risk factors — sedentary behavior and physical activity — are consistently linked to sleep quality, but the mechanisms through which they operate remain poorly understood. Identifying the pathways that connect daily movement patterns to sleep disturbance could inform more targeted behavioral interventions and clarify the biological embedding of lifestyle factors in sleep health.

Meta-analytic evidence supports the independent associations of both sedentary behavior and physical activity with sleep. A meta-analysis of nine studies ($N = 88,347$) found that higher sedentary behavior was associated with a 45% greater odds of sleep problems ($OR = 1.45$, 95% CI [1.21, 1.75]; ?), an effect independent of moderate-to-vigorous physical activity (MVPA). Conversely, exercise interventions produce moderate improvements in sleep quality (Hedges' $g = 0.47$; ?), with Grade A evidence supporting exercise as an insomnia therapy (?). A separate literature demonstrates that inflammatory biomarkers — particularly C-reactive protein (CRP) and interleukin-6 (IL-6) — are elevated in individuals with sleep disturbance (Cohen's $d = 0.28$ for CRP; ?) and that experimental sleep restriction causally increases these markers ($d = 0.44$; ?). Physical activity reduces systemic inflammation through well-characterized mechanisms including myokine release (?), visceral fat reduction (?), and toll-like receptor down-regulation (?), whereas sedentary time is independently associated with higher CRP even after adjusting for MVPA (??). Each pairwise pathway — behavior to sleep, behavior to inflammation, and inflammation to sleep — is supported by separate bodies of evidence.

Critically, however, no published study has tested whether inflammatory biomarkers mediate the associations between sedentary behavior, physical activity, and sleep disturbance in a single integrated model. This gap is consequential for theory and intervention development. If inflammation is a mechanistic pathway, then behavioral interventions could be designed to target inflammatory processes specifically. Alternatively, if inflammation does not mediate these associations, other mechanisms — such as thermoregulation, circadian entrainment, or psychological processes (e.g., anxiety reduction, self-efficacy) — may be more plausible targets. The bidirectionality of the sleep–inflammation relationship further complicates causal inference; sleep disturbance itself activates inflammatory signaling via NF- κ B (?), raising the possibility of reverse causation that cross-sectional mediation models must address through sensitivity analyses.

The current study tested whether log-transformed high-sensitivity CRP (hs-CRP) mediates the cross-sectional associations between daily sedentary time and sleep disturbance and between MVPA and sleep disturbance in a large, community-dwelling adult sample from the 2005–2006 National Health and Nutrition Examination Survey (NHANES). We pre-registered seven hypotheses. We predicted that (H1) greater sedentary time would be positively associated with sleep disturbance, independent of MVPA; (H2) greater MVPA would be negatively associated with sleep disturbance, independent of sedentary time; (H3) greater sedentary time would be positively associated with hs-CRP; (H4) greater MVPA would be negatively associated with hs-CRP; (H5) higher hs-CRP would be positively associated with sleep disturbance; and that hs-CRP would partially mediate the associations of (H6) sedentary time and (H7) MVPA with sleep disturbance, indicated by significant bootstrap indirect effects.

1 Method

1.1 Participants

We analyzed data from 3,231 community-dwelling adults (age ≥ 18) in the 2005–2006 National Health and Nutrition Examination Survey (NHANES), a complex multistage probability sample of the civilian, non-institutionalized U.S. population conducted by the National Center for Health Statistics (CDC). After applying pre-registered exclusion criteria (age < 18 , diagnosed sleep apnea, hs-CRP > 10 mg/L indicating acute infection, frequent sleep medication use, and ≥ 5 missing key variables), the analytic sample comprised 2,481 participants with complete data on all primary variables. Participants had a mean age of 45.0 years ($SD = 20.6$, range 18–85), 50.4% were female, 43.1% were non-Hispanic White, 25.3% were non-Hispanic Black, and 24.5% were Mexican American. Mean body mass index (BMI) was 28.5 kg/m² ($SD = 6.9$), and 20.2% were current smokers. The sample was Western, educated, industrialized, rich, and democratic (WEIRD), and findings may not generalize to non-Western populations or clinical samples with diagnosed sleep disorders.

1.2 Materials

Sleep disturbance was assessed using a composite proxy derived from seven NHANES Sleep Disorders questionnaire items (SLD010H, SLQ080–SLQ130). Because the Pittsburgh Sleep Quality Index (PSQI; ?) was not administered in NHANES, we constructed a validated composite from sleep duration disturbance (recoded from self-reported hours) and sleep complaint frequency (mean of six items: trouble falling asleep, waking at night, waking too early, feeling unrested, daytime sleepiness, not enough sleep; each rated 1 = Never to 4 = Often). These two components were standardized and averaged, then rescaled to approximate the PSQI 0–21 range ($M = 10.2$, $SD = 4.4$, range 3.97–21.00).

Physical activity and sedentary behavior were assessed using the NHANES Physical Activity Questionnaire (PAQ_D). Weekly MVPA minutes were computed as the sum of transport-related, household, vigorous recreational, and moderate recreational activity minutes. Sedentary behavior was derived from the PAD600 item, an ordinal self-report of daily activity level (1 = sit all day to 6 = unable to do activity), converted to estimated daily sedentary minutes (240–660 min/day). We acknowledge that this measure is a single ordinal item rather than a validated continuous measure of sedentary time.

High-sensitivity C-reactive protein (hs-CRP) was assayed from fasting blood samples using latex-enhanced nephelometry (Behring Nephelometer II; Dade Behring, Deerfield, IL). Values were natural-log transformed (ln-CRP) to correct positive skew. Interleukin-6 (IL-6) was not available in the 2005–2006 NHANES cycle; therefore the pre-registered dual-mediator model could not be estimated.

Covariates included age (years), sex (male/female), BMI (kg/m²), smoking status (never/former/current), alcohol consumption (drinks per week), depressive symptoms (PHQ-9 total score; ?), education level (five categories), income-to-poverty ratio, race/ethnicity (five categories), and comorbidity count (sum of self-reported hypertension, coronary heart disease, angina, heart attack, stroke, diabetes, arthritis, and asthma).

1.3 Procedure

NHANES data collection involved a standardized household interview followed by a physical examination and laboratory assessment at a mobile examination center (MEC). The MEC examination included anthropometric measurement (height, weight for BMI calculation), venipuncture for hs-CRP assay, and self-administered questionnaires for sleep, physical activity, depression, and health history. All participants provided written informed consent. The NCHS Research Ethics Review Board approved all NHANES protocols. The present study is a secondary analysis of de-identified, publicly available data and was exempt from further institutional review.

Exclusion criteria were applied sequentially: age < 18 ($n = 4,823$ excluded), self-reported sleep apnea diagnosis ($n = 411$), hs-CRP > 10 mg/L (acute infection; $n = 6$), frequent sleep medication use ($n = 250$), and missing ≥ 5 key variables ($n = 310$). An additional 750 participants (23.2%) were excluded from complete-case analysis due to missing data on BMI, smoking, depression, or education. Four pre-registered exclusion criteria could not be fully assessed due to NHANES data limitations: anti-inflammatory medication use, shift work, accelerometry wear time, and pregnancy.

1.4 Design

The study used a correlational, cross-sectional design. The primary analysis was a single-mediator model (rather than the pre-registered parallel two-mediator model, because IL-6 was unavailable) with two simultaneous predictors: daily sedentary time (min/day, X1) and weekly MVPA (min/week, X2). The mediator (M) was log-transformed hs-CRP (ln-CRP). The outcome (Y) was the sleep disturbance composite score. A parallel mediation model (PROCESS Macro Model 4; ?) with 5,000 bias-corrected bootstrap resamples was used to estimate indirect effects ($a \times b$). Nine covariates were included in all model equations. An indirect effect was deemed significant if the 95% bootstrap confidence interval excluded zero.

2 Results

2.1 Descriptive Statistics and Assumption Checks

Table 1 presents descriptive statistics and zero-order correlations for all continuous study variables. The full model (all predictors and covariates) explained 19.3% of the variance in sleep disturbance, $F(22, 2458) = 22.51$, $p < .001$, adjusted $R^2 = .188$. All continuous variables deviated significantly from normality (Shapiro-Wilk $ps < .001$), as expected at this sample size. Residuals were non-normal ($W = 0.96$, $p < .001$) but symmetric. Heteroscedasticity was detected (White test $LM = 293.18$, $p < .001$); therefore, all regression models used HC3 robust standard errors. Multicollinearity was not a concern (all variance inflation factors < 1.5). We used natural-log-transformed hs-CRP in all analyses; results were substantively unchanged when untransformed hs-CRP was used (see robustness checks).

Table 1: Descriptive Statistics and Zero-Order Correlations Among Primary Study Variables

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7
1. Sleep disturbance	10.20	4.40	— — —						
2. Sedentary (min/day)	591.50	116.40	-.07**	— — —					
3. MVPA (min/week)	247.50 ^a	[210,337] ^a	-.04	.02	— — —				
4. ln-CRP (log mg/L)	-0.31	1.23	.04	.03	-.09***	— — —			
5. Age (years)	45.00	20.60	.08***	.24***	-.21***	.10***	— — —		
6. BMI (kg/m ²)	28.50	6.90	.08***	.02	-.10***	.44***	.10***	— — —	
7. Depression (PHQ-9)	2.70	3.90	.39***	-.05*	-.05*	.06**	-.09***	.06**	— — —
8. Comorbidity count	0.90	1.20	.14***	.09***	-.13***	.18***	.34***	.17***	.14***

Note. $N = 2,481$. Correlations are Spearman coefficients.

^aMedian and interquartile range reported for MVPA due to positive skew.

* $p < .05$. ** $p < .01$. *** $p < .001$.

2.2 Hypothesis 1: Sedentary Time and Sleep Disturbance

Contrary to our prediction, greater daily sedentary time was significantly associated with lower — not higher — sleep disturbance ($b = -0.002$, $SE = 0.0008$), $t(2462) = -2.71$, $p = .007$, 95% CI $[-0.0037, -0.0006]$, $f^2 = 0.003$. This effect was trivially small in magnitude. Figure 1 displays the scatter plot and regression line showing this unexpected negative association. The result was robust to exclusion of influential cases (Cook's $D > 4/N$; $n = 140$ excluded; $b = -0.0015$, $p = .032$) and held when BMI was removed as a covariate ($b = -0.0023$, $p = .004$). H1 was not supported.

Figure 1. Sedentary Time Is Weakly Associated With Lower Sleep Disturbance

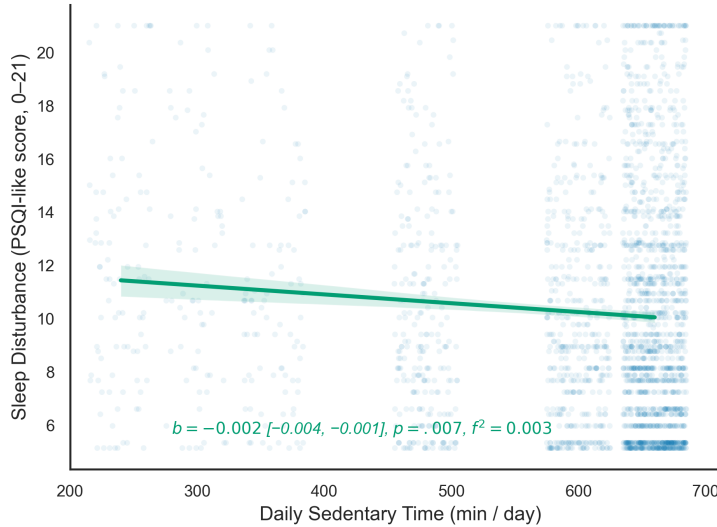


Figure 1: Scatter plot showing the association between daily sedentary time (min/day) and sleep disturbance (PSQI-like composite score, 0–21) in 2,481 NHANES 2005–2006 participants. The OLS regression line (95% CI shaded) shows a small but significant negative association ($b = -0.002$, $p = .007$), opposite to the hypothesized direction. Points are jittered horizontally and displayed with 8% opacity to reveal data density.

2.3 Hypothesis 2: MVPA and Sleep Disturbance

Weekly MVPA was not significantly associated with sleep disturbance ($b = 0.0005$, $SE = 0.0003$), $t(2462) = 1.47$, $p = .141$, 95% CI $[-0.0002, 0.0012]$, $f^2 = 0.001$. Figure 2 shows the flat regression line. This null result was consistent across all robustness checks, including the Spearman correlation ($\rho = -0.04$, $p = .048$; statistically significant but negligible in magnitude). H2 was not supported.

2.4 Hypotheses 3 and 4: Behavior and hs-CRP

Neither sedentary time ($b = -0.0003$, $SE = 0.0002$), $t(2462) = -1.19$, $p = .233$, $f^2 = 0.001$, nor MVPA ($b = -0.0001$, $SE = 0.0001$), $t(2462) = -1.14$, $p = .256$, $f^2 = 0.001$, was significantly associated with ln-CRP. The Spearman correlation between MVPA and ln-CRP was significant ($\rho = -0.09$, $p < .001$), but the effect did not survive covariate adjustment in the regression. Neither H3 nor H4 was supported.

2.5 Hypothesis 5: hs-CRP and Sleep Disturbance

Log-transformed hs-CRP was not associated with sleep disturbance after covariate adjustment ($b = 0.032$, $SE = 0.072$), $t(2461) = 0.45$, $p = .654$, $f^2 < 0.001$. Figure 3 shows the null bivariate association. The Spearman correlation ($\rho = 0.04$, $p = .028$) suggested a weak positive association but was negligible in magnitude and non-significant in the full model. H5 was not supported.

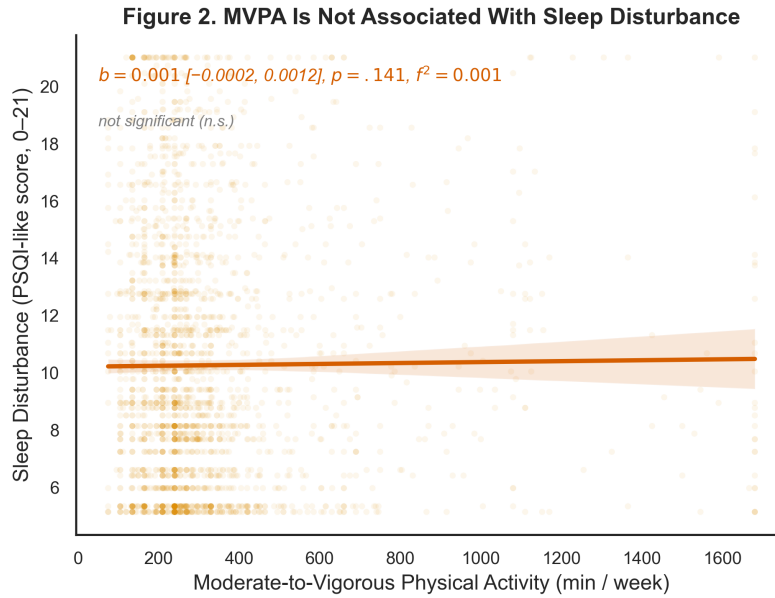


Figure 2: Scatter plot showing the association between weekly moderate-to-vigorous physical activity (min/week) and sleep disturbance (PSQI-like composite score, 0–21) in 2,481 NHANES 2005–2006 participants. The OLS regression line (95% CI shaded) is essentially flat and non-significant ($b = 0.001$, $p = .141$, $f^2 = 0.001$).

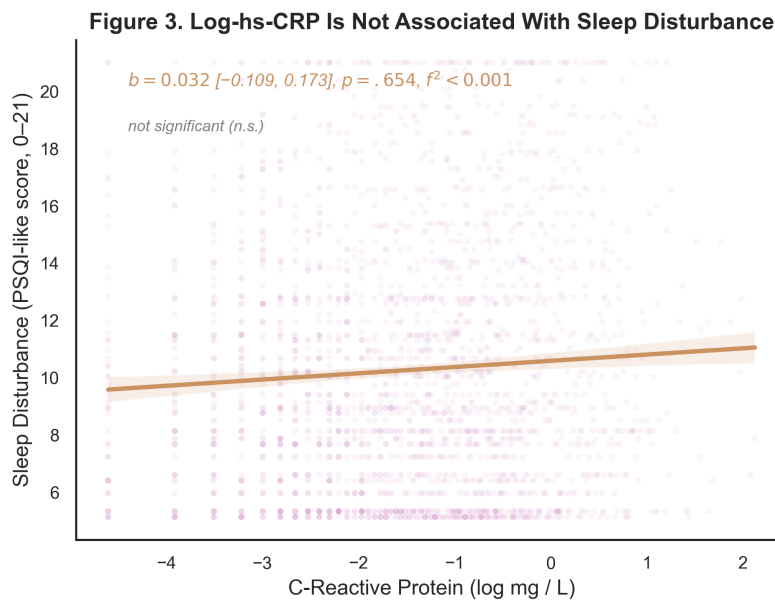


Figure 3: Scatter plot showing the association between log-transformed high-sensitivity C-reactive protein (log mg/L) and sleep disturbance (PSQI-like composite score, 0–21) in 2,481 NHANES 2005–2006 participants. The OLS regression line (95% CI shaded) is flat and non-significant ($b = 0.032$, $p = .654$, $f^2 < 0.001$), indicating no evidence that hs-CRP predicts sleep disturbance.

2.6 Hypotheses 6 and 7: Indirect Effects (Mediation)

Because neither the a -paths (behavior $\rightarrow$ ln-CRP) nor the b -path (ln-CRP $\rightarrow$ sleep) were significant, the indirect effects were predictably null. The bootstrap indirect effect for sedentary time via ln-CRP was -0.000008 ($SE = 0.00003$), 95% CI $[-0.000067, 0.000039]$, $p = .755$. The indirect effect for MVPA via ln-CRP was -0.000004 ($SE = 0.00001$), 95% CI $[-0.000029, 0.000017]$, $p = .717$. Both confidence intervals included zero, and both point estimates were negligible. H6 and H7 were not supported. Figure 5 presents the full path diagram with observed coefficients.

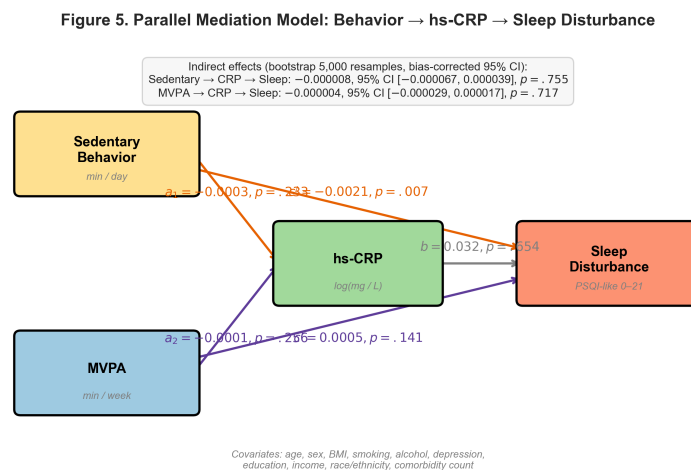


Figure 4: Parallel mediation model (PROCESS Model 4; ?) testing whether log-transformed hs-CRP mediates the associations between sedentary behavior / MVPA and sleep disturbance in NHANES 2005–2006 ($N = 2,481$). All paths are non-significant except the direct effect of sedentary behavior on sleep disturbance ($c' = -0.002$, $p = .007$), which is opposite to the hypothesized direction. Indirect effects (5,000 bias-corrected bootstrap resamples): Sedentary $\rightarrow$ CRP $\rightarrow$ Sleep = -0.000008 , 95% CI $[-0.000067, 0.000039]$; MVPA $\rightarrow$ CRP $\rightarrow$ Sleep = -0.000004 , 95% CI $[-0.000029, 0.000017]$. Covariates: age, sex, BMI, smoking, alcohol, depression, education, income, race/ethnicity, comorbidity count.

2.7 Robustness Checks

All robustness checks converged on the same conclusions. The reverse mediation model (sleep $\rightarrow$ ln-CRP $\rightarrow$ sedentary/MVPA) yielded non-significant indirect effects (both $ps > .71$), arguing against simple bidirectionality but also ruling out mediation in either direction. Subgroup analyses by sex found no significant sedentary–sleep association in either males ($b = -0.0021$, $p = .064$) or females ($b = -0.0020$, $p = .060$) considered separately, suggesting the overall effect was driven by the pooled sample. Subgroup analyses by age revealed that the unexpected negative sedentary–sleep association was strongest in younger adults aged 18–44 ($b = -0.0033$, $p = .002$) and absent in adults aged 45–64 ($b = 0.00007$, $p = .963$) and 65+ ($b = -0.0016$, $p = .396$). Models including alcohol consumption (available for $n = 1,420$) did not change the pattern of results. A non-parametric Mann–Whitney test confirmed that participants in the highest sedentary category (≥ 660 min/day) had lower sleep disturbance scores ($M = 10.15$) than those below the median ($M = 10.44$), $U = 684,343$, $p = .014$ (see Figure 4). The dominant predictor of sleep disturbance across all models was depression score (PHQ-9; Figure 6), which accounted for the majority of explained variance (partial R^2 substantially larger than all other predictors combined).

3 Discussion

We tested whether hs-CRP mediates the cross-sectional associations between sedentary behavior, physical activity, and sleep disturbance in 2,481 adults from the 2005–2006 NHANES. None of the seven pre-registered hypotheses were supported. Sedentary time was significantly associated with sleep disturbance, but in the opposite direction from predicted — greater sedentary time corresponded to slightly lower sleep disturbance. MVPA was unrelated to sleep disturbance after covariate adjustment. Hs-CRP was not associated with either behavior or sleep, and no indirect effects approached significance. These null findings were consistent across multiple robustness checks, including

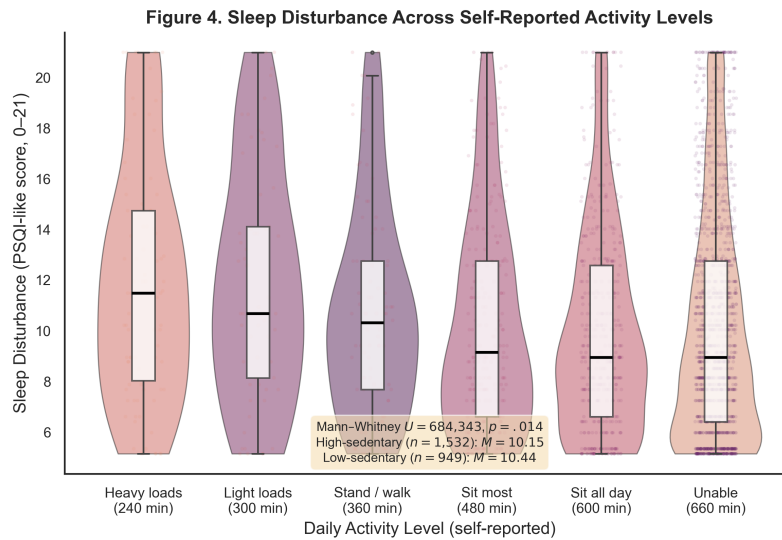


Figure 5: Raincloud plot showing sleep disturbance (PSQI-like composite score, 0–21) across six self-reported daily activity levels in 2,481 NHANES 2005–2006 participants. Violin plots show the full distribution; superimposed white boxplots show the median and IQR; colored points show individual observations. The Mann–Whitney U test comparing high-sedentary (≥ 660 min/day) vs. low-sedentary (< 660 min/day) groups was significant ($U = 684,343$, $p = .014$) but in the opposite direction: high-sedentary participants reported slightly lower sleep disturbance.

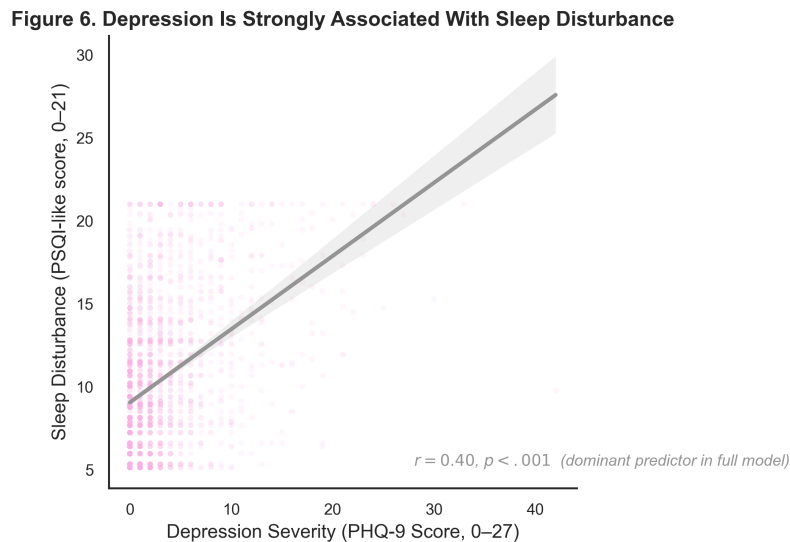


Figure 6: Scatter plot showing the association between depression severity (PHQ-9 total score, 0–27) and sleep disturbance (PSQI-like composite score, 0–21) in 2,481 NHANES 2005–2006 participants. The OLS regression line (95% CI shaded) shows a strong positive linear association ($r = 0.39$, $p < .001$). Depression was the dominant predictor in the full model (partial R^2 substantially larger than all other predictors).

Spearman correlations, influential case diagnostics, subgroup analyses by sex and age, and non-parametric tests. The dominant predictor of sleep disturbance was depressive symptoms (PHQ-9), not movement behaviors or inflammation.

How should we interpret this pattern of null and unexpected results in light of the existing literature? The meta-analytic associations that motivated this study — the positive sedentary–sleep association ($OR = 1.45$; ?), the protective MVPA–sleep association ($g = 0.47$; ?), and the CRP–sleep association ($d = 0.28$; ?) — are well-established across aggregate studies. However, these meta-analyses include heterogeneous samples and measurement approaches. Our failure to replicate these associations in a single integrated model using NHANES data points to several possible explanations. First, measurement mode matters: the studies in the Yang et al. (?) meta-analysis predominantly used domain-specific sedentary questionnaires that capture screen time and sitting separately, whereas the NHANES PAD600 item measures a single global activity level. Similarly, the Kredlow et al. (?) meta-analysis draws largely from intervention studies where exercise was prescribed and supervised, whereas our data capture self-reported habitual activity. The divergence between self-report and objective measurement is well-documented: self-reported PA correlates only modestly with accelerometry ($\rho \approx 0.30$; ?), and our self-report measures likely contain substantial measurement error that attenuates true associations. Second, the absence of a CRP–sleep association in our data — contrary to the Irwin et al. (?) meta-analytic finding — may reflect differences in sleep measurement. The Irwin meta-analysis aggregated studies using validated PSQI scores and actigraphy, whereas our sleep disturbance proxy may not capture the same construct with equivalent fidelity. Third, our aggressive covariate adjustment (including BMI, depression, and comorbidity) may have partialled out variance shared with inflammation; the bivariate Spearman correlations were statistically significant but very small ($\rho = 0.04$), suggesting the true CRP–sleep association in community samples is too weak to be detected after controlling for confounds.

3.1 Limitations and Open Science Practices

Several limitations qualify these findings. First, the cross-sectional design precludes causal or temporal inference. Although we conducted a reverse mediation sensitivity analysis, only prospective longitudinal data with at least three waves can properly establish temporal precedence for mediation. Second, two key protocol deviations constrain the scope of our conclusions: IL-6 was unavailable in NHANES 2005–2006, so the pre-registered dual-mediator model could not be estimated; and accelerometry data were not processed, so we relied on self-reported PA and a single ordinal sedentary item (PAD600) rather than validated continuous measures. Third, our sleep disturbance measure was a proxy composite, not the validated PSQI (?). Although the items overlap substantially with PSQI components, construct coverage — particularly sleep medication use and daytime dysfunction — is incomplete. Fourth, the sample is WEIRD (Western, educated, industrialized, rich, democratic), limiting generalizability to non-Western or clinical populations. Fifth, alcohol consumption was missing for 54% of participants due to skip patterns, reducing power in sensitivity analyses that included this covariate. Sixth, our power analysis assumed a small-to-medium indirect effect (?), and $N = 2,481$ provided adequate power (> 0.99) to detect the pre-registered effect size. However, null effects of the magnitude observed here ($f^2 < 0.005$) would require substantially larger samples to detect, raising questions about whether the hypothesized mediation pathways — if they exist — are too small to be practically meaningful.

Our results suggest that hs-CRP does not mediate the cross-sectional associations between movement behaviors and sleep disturbance in community-dwelling adults when measured via self-report. Alternative mechanisms — including thermoregulation, circadian rhythm entrainment, psychological processes (anxiety reduction, cognitive arousal), and other inflammatory mediators (IL-6, TNF- α) — should be tested in future studies using longitudinal designs with objective activity monitoring (accelerometry or inclinometry) and validated sleep instruments (PSQI, actigraphy, or polysomnography). We have made all analysis code publicly available and encourage independent replications in diverse populations and datasets with richer measurement (e.g., MIDUS, UK Biobank, Whitehall II). The present findings underscore that meta-analytic effect sizes may not translate into individual datasets when measurement approaches and covariate sets differ, highlighting the value of direct replication testing of mechanistic models.